Problem Bank to accompany Homework 2. Each item is a short, exam-style question targeting one specific idea from the homework (excluding the Lin-estimator simulation). The emphasis is on *computation*: hand-inversion of 2×2 matrices, explicit construction of projection and annihilator matrices, and numerical verification of identities that only become intuitive once you've cranked out the arithmetic. Try each question on your own before consulting the solutions, which are collected at the end.

Concept A: Projection matrices and bivariate OLS by hand

A.1 Consider the data

$$x = (1, 2, 4)', \qquad y = (3, 5, 9)'.$$

Let $\mathbf{X} = [\mathbf{1}\ \mathbf{x}]$ (intercept plus slope).

(a) Compute $\mathbf{X}'\mathbf{X}$ explicitly as a 2×2 matrix.

(b) Compute $(\mathbf{X}'\mathbf{X})^{-1}$ by the 2×2 inverse formula.

(c) Compute $\mathbf{X}'\mathbf{y}$ and then $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

(d) Report the OLS fitted values $\hat{\mathbf{y}}$ and residuals $\hat{\mathbf{e}}$. Verify $\mathbf{1}'\hat{\mathbf{e}} = 0$ and $\mathbf{x}'\hat{\mathbf{e}} = 0$.

A.2 Continuing with the data from A.1, compute the projection matrix

$$\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

explicitly as a 3×3 matrix. Then:

(a) Verify $\mathbf{P}\mathbf{1} = \mathbf{1}$ and $\mathbf{P}\mathbf{x} = \mathbf{x}$ (the two columns of $\mathbf{X}$ are fixed points).

(b) Verify $\mathbf{P}^2 = \mathbf{P}$ by computing one entry explicitly.

(c) Compute $\operatorname{tr}(\mathbf{P})$. What does this number equal, and why?

A.3 Using the same data, verify the identity

$$\sum_{i=1}^{n} x_i^2 - n\bar{x}^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2$$

by computing both sides explicitly. Then use this identity to simplify the formula

$$\hat{\beta}_1 = \frac{n\sum x_i y_i - (\sum x_i)(\sum y_i)}{n\sum x_i^2 - (\sum x_i)^2}$$

to $\hat{\beta}_1 = \sum(x_i - \bar{x})(y_i - \bar{y})/\sum(x_i - \bar{x})^2$, and compute $\hat{\beta}_1$ directly in this form. Confirm it agrees with A.1(c).

A.4 FWL by hand. Using the same data, compute $\hat{\beta}_1$ by demeaning (the Frisch–Waugh–Lovell recipe):

(a) Compute $\bar{x} = 7/3$ and $\bar{y} = 17/3$. Form the demeaned vectors $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$.

(b) Run OLS of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{x}}$ (no intercept): $\hat{\beta}_1 = (\tilde{\mathbf{x}}'\tilde{\mathbf{x}})^{-1}\tilde{\mathbf{x}}'\tilde{\mathbf{y}}$.

(c) Confirm you recover the same $\hat{\beta}_1$ as in A.1 and A.3.

A.5 FWL with a second regressor. Consider the data

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 4 & 6 \end{bmatrix}, \qquad \mathbf{y} = (2, 4, 7, 8)',$$

where the columns are intercept, $\mathbf{x}_1$, $\mathbf{x}_2$.

(a) Regress $\mathbf{x}_2$ on $[\mathbf{1}\ \mathbf{x}_1]$ and compute the residual vector $\tilde{\mathbf{x}}_2$.

(b) Regress $\mathbf{y}$ on $[\mathbf{1}\ \mathbf{x}_1]$ and compute the residual vector $\tilde{\mathbf{y}}$.

(c) Compute the FWL coefficient $\hat{\beta}_2 = (\tilde{\mathbf{x}}_2'\tilde{\mathbf{x}}_2)^{-1}\tilde{\mathbf{x}}_2'\tilde{\mathbf{y}}$.

Concept B: Group fixed effects and the within estimator

B.1 Consider a panel with $N = 5$ individuals grouped into $J = 2$ districts, with $n_1 = 2$ and $n_2 = 3$. The group-membership matrix is

$$\mathbf{M}_J = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

(a) Compute $\mathbf{M}_J'\mathbf{M}_J$. What is its general form for J groups?

(b) Compute $(\mathbf{M}_J'\mathbf{M}_J)^{-1}$.

B.2 Using the $\mathbf{M}_J$ from B.1, compute the projection matrix $\mathbf{M}_J(\mathbf{M}_J'\mathbf{M}_J)^{-1}\mathbf{M}_J'$ as an explicit 5×5 matrix. Identify the block-diagonal structure with $(1/n_j)\mathbf{J}_{n_j}$ blocks.

B.3 A researcher has the following observations (two groups, one regressor x and one outcome y):

Group	y	x	unit
1	4	2	1
1	6	4	2
2	10	5	3
2	14	7	4
2	12	6	5

(a) Compute the group means $\bar{y}_j$ and $\bar{x}_j$ for $j = 1, 2$.

(b) Compute the within-transformed data $\tilde{y}_i = y_i - \bar{y}_{j(i)}$ and $\tilde{x}_i = x_i - \bar{x}_{j(i)}$ for all five observations.

B.4 Continuing with B.3, compute the within estimator

$$\hat{\beta} = \frac{\sum_i \tilde{x}_i\,\tilde{y}_i}{\sum_i \tilde{x}_i^2}.$$

Then predict y_i for each observation using $\hat{y}_i = \hat{\alpha}_j + \hat{\beta}x_i$, where $\hat{\alpha}_j = \bar{y}_j - \hat{\beta}\bar{x}_j$. Report residuals and verify they sum to zero *within each group*.

B.5 Consider the single-district case $J = 1$ (all observations in one group).

(a) What does $\mathbf{M}_J$ become? What does the within transformation reduce to?

(b) Show that the within estimator coincides with the ordinary OLS estimator of y on a constant and x.

Concept C: Recursive OLS and rank-one updates

Throughout, let $\mathbf{X}_n$ be $n \times k$ and $\hat{\beta}_n = (\mathbf{X}_n'\mathbf{X}_n)^{-1}\mathbf{X}_n'\mathbf{y}_n$. When a new observation $(y_{n+1}, \mathbf{x}_{n+1})$ arrives,

$$\mathbf{X}_{n+1}'\mathbf{X}_{n+1} = \mathbf{X}_n'\mathbf{X}_n + \mathbf{x}_{n+1}\mathbf{x}_{n+1}'.$$

The Sherman–Morrison formula states

$$(\mathbf{A} + \mathbf{u}\mathbf{u}')^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'\mathbf{A}^{-1}}{1 + \mathbf{u}'\mathbf{A}^{-1}\mathbf{u}}.$$

C.1 Let $\mathbf{A} = \mathbf{X}_n'\mathbf{X}_n = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$.

(a) Compute $\mathbf{A}^{-1}$ by the 2×2 inverse formula.

(b) A new observation arrives with $\mathbf{x}_{n+1} = (1, 2)'$. Compute $\mathbf{X}_{n+1}'\mathbf{X}_{n+1} = \mathbf{A} + \mathbf{x}_{n+1}\mathbf{x}_{n+1}'$ explicitly.

(c) Compute $(\mathbf{X}_{n+1}'\mathbf{X}_{n+1})^{-1}$ directly using the 2×2 inverse formula.

C.2 Use the Sherman–Morrison formula to compute $(\mathbf{X}_{n+1}'\mathbf{X}_{n+1})^{-1}$ from the setup in C.1, and verify it matches your answer to C.1(c).

(a) Compute the scalar $1 + \mathbf{x}_{n+1}'\mathbf{A}^{-1}\mathbf{x}_{n+1}$.

(b) Compute the rank-one update $\mathbf{A}^{-1}\mathbf{x}_{n+1}\mathbf{x}_{n+1}'\mathbf{A}^{-1}$ as a 2×2 matrix.

(c) Subtract and report the result.

C.3 Continuing with the same setup, suppose $\hat{\beta}_n = (1, 0.5)'$ and the new observation is $(y_{n+1}, \mathbf{x}_{n+1}) = (3, (1, 2)')$. Use the recursive formula

$$\hat{\beta}_{n+1} = \hat{\beta}_n + \frac{(\mathbf{X}_n'\mathbf{X}_n)^{-1}\mathbf{x}_{n+1}\,(y_{n+1} - \mathbf{x}_{n+1}'\hat{\beta}_n)}{1 + \mathbf{x}_{n+1}'(\mathbf{X}_n'\mathbf{X}_n)^{-1}\mathbf{x}_{n+1}}$$

to compute $\hat{\beta}_{n+1}$.

C.4 The recursive formula has an appealing structure: $\hat{\beta}_{n+1}$ is $\hat{\beta}_n$ plus a correction proportional to the prediction error on the new observation.

(a) What is the "gain" (scalar multiplier on the prediction error) when $\mathbf{x}_{n+1}$ lies deep in the column space of prior $\mathbf{X}_n$ (i.e., $\mathbf{x}_{n+1}'(\mathbf{X}_n'\mathbf{X}_n)^{-1}\mathbf{x}_{n+1}$ is small)?

(b) What is it when $\mathbf{x}_{n+1}$ is an extreme point (this quantity is large)?

(c) Interpret: when does the new observation strongly move the estimate?

C.5 Leverage limit. The quantity $h_{n+1,n+1} = \mathbf{x}'_{n+1}(\mathbf{X}'_{n+1}\mathbf{X}_{n+1})^{-1}\mathbf{x}_{n+1}$ is the leverage of the new observation in the augmented sample. Prove that

$$h_{n+1,n+1} = \frac{\mathbf{x}'_{n+1}(\mathbf{X}'_n\mathbf{X}_n)^{-1}\mathbf{x}_{n+1}}{1 + \mathbf{x}'_{n+1}(\mathbf{X}'_n\mathbf{X}_n)^{-1}\mathbf{x}_{n+1}},$$

using Sherman–Morrison. Deduce $h_{n+1,n+1} \in [0, 1)$ regardless of how extreme the new observation is.

Solutions

A.1. $\sum x_i = 7$, $\sum x_i^2 = 1+4+16 = 21$, $\sum y_i = 17$, $\sum x_i y_i = 3+10+36 = 49$. (a) $\mathbf{X}'\mathbf{X} = \begin{bmatrix} 3 & 7 \\ 7 & 21 \end{bmatrix}$. (b) det $= 63 - 49 = 14$, so

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{14}\begin{bmatrix} 21 & -7 \\ -7 & 3 \end{bmatrix} = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 3/14 \end{bmatrix}.$$

(c) $\mathbf{X}'\mathbf{y} = (17, 49)'$. $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$: $\hat{\beta}_0 = (3/2)(17) + (-1/2)(49) = 25.5 - 24.5 = 1$. $\hat{\beta}_1 = (-1/2)(17) + (3/14)(49) = -8.5 + 10.5 = 2$. So $\hat{\boldsymbol{\beta}} = (1,2)'$. (d) Fitted values $\hat{y}_i = 1 + 2x_i$: $\hat{\mathbf{y}} = (3,5,9)'$. Residuals: $\hat{\mathbf{e}} = (0,0,0)'$. The three points lie exactly on the line $y = 1+2x$, so residuals and both orthogonality conditions hold trivially.

A.2. Let $\mathbf{x}_i$ denote the ith row of $\mathbf{X}$: $\mathbf{x}_1 = (1,1)$, $\mathbf{x}_2 = (1,2)$, $\mathbf{x}_3 = (1,4)$. Entry $P_{ij} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_j$. Using $(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{14}\begin{bmatrix} 21 & -7 \\ -7 & 3 \end{bmatrix}$ and computing each entry:

$$\mathbf{P} = \frac{1}{14}\begin{bmatrix} 10 & 6 & -2 \\ 6 & 5 & 3 \\ -2 & 3 & 13 \end{bmatrix}.$$

Check of P_{11}: $\mathbf{x}_1'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_1 = (1/14)[1,1]\begin{bmatrix} 14 \\ -4 \end{bmatrix} = 10/14$. ✓. (a) Row sums of $\mathbf{P}$ all equal $14/14 = 1$, so $\mathbf{P1} = \mathbf{1}$. $\mathbf{Px}$: row 1 gives $(10+12-8)/14 = 1$; row 2 gives $(6+10+12)/14 = 2$; row 3 gives $(-2+6+52)/14 = 4$. So $\mathbf{Px} = (1,2,4)' = \mathbf{x}$. (b) Idempotency (check one entry): $(\mathbf{P}^2)_{11} = (1/14^2)[10\cdot 10 + 6\cdot 6 + (-2)\cdot(-2)] = (100+36+4)/196 = 140/196 = 10/14$. ✓(c) $\mathrm{tr}(\mathbf{P}) = (10+5+13)/14 = 28/14 = 2$. Equals $\mathrm{rank}(\mathbf{X}) = 2$, the dimension of the column space onto which we project.

A.3. $\bar{x} = 7/3$, $n\bar{x}^2 = 3(49/9) = 49/3$. $\sum x_i^2 = 21 = 63/3$. LHS: $63/3 - 49/3 = 14/3$. RHS: $(1-7/3)^2 + (2-7/3)^2 + (4-7/3)^2 = (-4/3)^2 + (-1/3)^2 + (5/3)^2 = (16+1+25)/9 = 42/9 = 14/3$. ✓Similarly $\sum(x_i - \bar{x})(y_i - \bar{y})$: $\bar{y} = 17/3$, so $y - \bar{y} = (-8/3, -2/3, 10/3)$. Product-sum: $(-4/3)(-8/3) + (-1/3)(-2/3) + (5/3)(10/3) = (32+2+50)/9 = 84/9 = 28/3$. $\hat{\beta}_1 = (28/3)/(14/3) = 2$, matching A.1.

A.4. (a) $\bar{x} = 7/3, \bar{y} = 17/3$. $\tilde{\mathbf{x}} = (-4/3, -1/3, 5/3)'$, $\tilde{\mathbf{y}} = (-8/3, -2/3, 10/3)'$. (b) $\tilde{\mathbf{x}}'\tilde{\mathbf{x}} = 14/3$ (from A.3), $\tilde{\mathbf{x}}'\tilde{\mathbf{y}} = 28/3$ (from A.3), so $\hat{\beta}_1 = 28/14 = 2$. (c) Matches. Intercept recovered as $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 17/3 - 2(7/3) = 3/3 = 1$.

A.5. First stage: regress $\mathbf{x}_2 = (2,3,5,6)'$ on $[\mathbf{1}\ \mathbf{x}_1]$ with $\mathbf{x}_1 = (1,2,3,4)'$. $\bar{x}_1 = 2.5, \bar{x}_2 = 4$. $S_{x_1x_1} = \sum(x_1 - \bar{x}_1)^2 = 2.25 + 0.25 + 0.25 + 2.25 = 5$. $S_{x_1x_2} = (1-2.5)(-2) + (2-2.5)(-1) + (3-2.5)(1) + (4-2.5)(2) = 3 + 0.5 + 0.5 + 3 = 7$. So slope of x_2 on x_1 is $7/5 = 1.4$; intercept $4 - 1.4(2.5) = 0.5$. Residuals $\tilde{x}_2 = x_2 - 0.5 - 1.4x_1 = (2-1.9, 3-3.3, 5-4.7, 6-6.1) = (0.1, -0.3, 0.3, -0.1)$. Similarly residualize $\mathbf{y} = (2,4,7,8)'$: $\bar{y} = 21/4 = 5.25$, $S_{x_1y} = (1-2.5)(2-5.25) + (2-2.5)(4-5.25) + (3-2.5)(7-5.25) + (4-2.5)(8-5.25) = 4.875 + 0.625 + 0.875 + 4.125 = 10.5$. Slope $10.5/5 = 2.1$, intercept $5.25 - 2.1(2.5) = 0$. Residuals: $\tilde{y} = y - 0 - 2.1x_1 = (2-2.1, 4-4.2, 7-6.3, 8-8.4) = (-0.1, -0.2, 0.7, -0.4)$. FWL slope: $\hat{\beta}_2 = \tilde{x}_2'\tilde{y}/\tilde{x}_2'\tilde{x}_2 = ((0.1)(-0.1) + (-0.3)(-0.2) + (0.3)(0.7) + (-0.1)(-0.4))/(0.01 + 0.09 + 0.09 + 0.01) = (-0.01 + 0.06 + 0.21 + 0.04)/0.20 = 0.30/0.20 = 1.5$.

B.1. (a) $\mathbf{M}_J'\mathbf{M}_J = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = \text{diag}(n_1, n_2)$. In general $\mathbf{M}_J'\mathbf{M}_J = \text{diag}(n_1, \ldots, n_J)$ because column j of $\mathbf{M}_J$ is ι_{n_j} and distinct columns have disjoint supports. (b) $(\mathbf{M}_J'\mathbf{M}_J)^{-1} = \text{diag}(1/n_1, 1/n_2) = \text{diag}(1/2, 1/3)$.

B.2. $\mathbf{M}_J(\mathbf{M}_J'\mathbf{M}_J)^{-1}\mathbf{M}_J'$ is a 5×5 matrix. Using the diagonal inverse, each column j of $\mathbf{M}_J$ contributes $(1/n_j)\iota_{n_j}\iota_{n_j}' = (1/n_j)\mathbf{J}_{n_j}$ in a block. Explicit form:

$$\mathbf{M}_J(\mathbf{M}_J'\mathbf{M}_J)^{-1}\mathbf{M}_J' = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 1/3 & 1/3 & 1/3 \end{bmatrix}.$$

Applied to a vector $\mathbf{y}$, this matrix returns a vector whose ith entry is $\bar{y}_{j(i)}$ — the group mean of i's group, replicated across rows.

B.3. (a) Group 1: $\bar{y}_1 = 5, \bar{x}_1 = 3$. Group 2: $\bar{y}_2 = 12, \bar{x}_2 = 6$. (b) Within-transformed:

Unit	$\tilde{y}$	$\tilde{x}$
1	$4 - 5 = -1$	$2 - 3 = -1$
2	$6 - 5 = 1$	$4 - 3 = 1$
3	$10 - 12 = -2$	$5 - 6 = -1$
4	$14 - 12 = 2$	$7 - 6 = 1$
5	$12 - 12 = 0$	$6 - 6 = 0$

B.4. $\sum \tilde{x}_i\tilde{y}_i = (-1)(-1)+(1)(1)+(-1)(-2)+(1)(2)+0 = 1+1+2+2+0 = 6$. $\sum \tilde{x}_i^2 = 1+1+1+1+0 = 4$. $\hat{\beta} = 6/4 = 1.5$. Group intercepts: $\hat{\alpha}_1 = 5 - 1.5 \cdot 3 = 0.5$; $\hat{\alpha}_2 = 12 - 1.5 \cdot 6 = 3$. Fitted:

Unit	x	y	$\hat{y}$	$\hat{e}$
1	2	4	$0.5 + 3 = 3.5$	0.5
2	4	6	$0.5 + 6 = 6.5$	-0.5
3	5	10	$3 + 7.5 = 10.5$	-0.5
4	7	14	$3 + 10.5 = 13.5$	0.5
5	6	12	$3 + 9 = 12$	0

Group-1 residuals sum to $0.5 - 0.5 = 0$; group-2 sum to $-0.5 + 0.5 + 0 = 0$. The within estimator orthogonalizes residuals within each group, as required by the normal equations for group dummies.

B.5. (a) $\mathbf{M}_J = \iota$ (the $N \times 1$ vector of ones), and the within transformation $\tilde{y}_i = y_i - \bar{y}$ becomes ordinary centering around the grand mean. (b) $\hat{\beta}_{\text{within}} = \sum(x_i - \bar{x})(y_i - \bar{y}) / \sum(x_i - \bar{x})^2$, which is the standard bivariate OLS slope. With $J = 1$ there are no "group dummies" beyond the single intercept, so the within transformation is just centering and recovers ordinary OLS.

C.1. (a) $\det \mathbf{A} = 12 - 4 = 8$, so $\mathbf{A}^{-1} = \frac{1}{8}\begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 3/8 & -1/4 \\ -1/4 & 1/2 \end{bmatrix}$. (b) $\mathbf{x}_{n+1}\mathbf{x}_{n+1}' = \begin{bmatrix} 1 \\ 2 \end{bmatrix}\begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. So $\mathbf{X}_{n+1}'\mathbf{X}_{n+1} = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix}$. (c) $\det = 35 - 16 = 19$, so $(\mathbf{X}_{n+1}'\mathbf{X}_{n+1})^{-1} = \frac{1}{19}\begin{bmatrix} 7 & -4 \\ -4 & 5 \end{bmatrix}$.

C.2. (a) $\mathbf{A}^{-1}\mathbf{x}_{n+1} = \begin{bmatrix} 3/8 & -1/4 \\ -1/4 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3/8 - 1/2 \\ -1/4 + 1 \end{bmatrix} = \begin{bmatrix} -1/8 \\ 3/4 \end{bmatrix}$. $\mathbf{x}'_{n+1}\mathbf{A}^{-1}\mathbf{x}_{n+1} = (1)(-1/8) + (2)(3/4) = -1/8 + 3/2 = 11/8$. So $1 + 11/8 = 19/8$. (b) $\mathbf{A}^{-1}\mathbf{x}_{n+1}\mathbf{x}'_{n+1}\mathbf{A}^{-1} = (\mathbf{A}^{-1}\mathbf{x}_{n+1})(\mathbf{A}^{-1}\mathbf{x}_{n+1})' = \begin{bmatrix} -1/8 \\ 3/4 \end{bmatrix} \begin{bmatrix} -1/8 & 3/4 \end{bmatrix} = \begin{bmatrix} 1/64 & -3/32 \\ -3/32 & 9/16 \end{bmatrix}$. (c) $\mathbf{A}^{-1} - \frac{1}{19/8} \cdot (\text{above}) = \mathbf{A}^{-1} - \frac{8}{19} \begin{bmatrix} 1/64 & -3/32 \\ -3/32 & 9/16 \end{bmatrix} = \mathbf{A}^{-1} - \begin{bmatrix} 1/152 & -3/76 \\ -3/76 & 9/38 \end{bmatrix}$. Writing $\mathbf{A}^{-1}$ with denominator 152: $\begin{bmatrix} 57/152 & -38/152 \\ -38/152 & 76/152 \end{bmatrix}$. Subtracting: $\begin{bmatrix} 56/152 & -32/152 \\ -32/152 & 40/152 \end{bmatrix} = \frac{1}{19} \begin{bmatrix} 7 & -4 \\ -4 & 5 \end{bmatrix}$. Matches C.1(c). ✓

C.3. Prediction error: $y_{n+1} - \mathbf{x}'_{n+1}\hat{\beta}_n = 3 - (1 \cdot 1 + 2 \cdot 0.5) = 3 - 2 = 1$. Gain denominator: $1 + \mathbf{x}'_{n+1}\mathbf{A}^{-1}\mathbf{x}_{n+1} = 19/8$. Numerator vector: $\mathbf{A}^{-1}\mathbf{x}_{n+1} = (-1/8, 3/4)'$ (from C.2). Update:

$$\hat{\beta}_{n+1} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + \frac{1}{19/8} \begin{bmatrix} -1/8 \\ 3/4 \end{bmatrix} \cdot 1 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + \frac{8}{19} \begin{bmatrix} -1/8 \\ 3/4 \end{bmatrix} = \begin{bmatrix} 1 - 1/19 \\ 0.5 + 6/19 \end{bmatrix} = \begin{bmatrix} 18/19 \\ 15.5/19 \end{bmatrix} \approx \begin{bmatrix} 0.947 \\ 0.816 \end{bmatrix}.$$

C.4. The gain is $[1 + \mathbf{x}'_{n+1}\mathbf{A}^{-1}\mathbf{x}_{n+1}]^{-1}\mathbf{A}^{-1}\mathbf{x}_{n+1}$, where the scalar denominator is one plus a quadratic form in $\mathbf{A}^{-1}$. (a) When $\mathbf{x}'_{n+1}\mathbf{A}^{-1}\mathbf{x}_{n+1}$ is small (observation is in a well-sampled part of the design), the denominator is close to 1 and the gain is close to $\mathbf{A}^{-1}\mathbf{x}_{n+1}$ — a routine update. (b) When the quadratic form is large (extreme / high-leverage observation), the denominator grows with it and the gain shrinks toward 0. (c) Interpretation: *a single extreme observation does not hijack the estimate.* The Sherman–Morrison denominator automatically shrinks the update to prevent a wildly-leveraged point from dominating. This is the algebraic reason leverage values h_{ii} are bounded below 1 (see C.5).

C.5. Let $\mathbf{B} = \mathbf{X}'_n\mathbf{X}_n$, $\mathbf{u} = \mathbf{x}_{n+1}$, and $q = \mathbf{u}'\mathbf{B}^{-1}\mathbf{u}$. By Sherman–Morrison,

$$(\mathbf{B} + \mathbf{u}\mathbf{u}')^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1}\mathbf{u}\mathbf{u}'\mathbf{B}^{-1}}{1 + q}.$$

Then $h_{n+1,n+1} = \mathbf{u}'(\mathbf{B} + \mathbf{u}\mathbf{u}')^{-1}\mathbf{u} = \mathbf{u}'\mathbf{B}^{-1}\mathbf{u} - \frac{(\mathbf{u}'\mathbf{B}^{-1}\mathbf{u})^2}{1+q} = q - \frac{q^2}{1+q} = \frac{q(1+q)-q^2}{1+q} = \frac{q}{1+q}$. Since $\mathbf{B}$ is positive definite, $q \geq 0$, so $h_{n+1,n+1} = q/(1+q) \in [0, 1)$. As $q \to \infty$, leverage approaches (but never reaches) 1. This bound is the formal reason the Sherman–Morrison update stays well-defined no matter how extreme the new observation is.